Biden Administration Bioeconomy Bold Goals Report

PLASTICS Response and Recommendations

October 5, 2023

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Introduction

On September 12, 2022, President Biden signed an Executive Order (EO) on "Advancing Biotechnology and Biomanufacturing Innovation for a Sustainable, Safe, and Secure American Bioeconomy". Ultimately, the intention behind this EO is to increase investments into the bioeconomy to "create new jobs, build stronger supply chains, and contribute to our climate goals[1]". To elaborate on what theoretically could be done to accomplish this intent, the *Bold Goals for US Biotechnology and Biomanufacturing* (Bold Goals) was published by the Department of Energy (DOE), Department of Agriculture (USDA), Department of Commerce (DOC), Department of Health and Human Services (HHS), and National Science Foundation (NSF), with input from other Federal departments and agencies.

The Bold Goals list potential action items that each of these contributing agencies could work on in conjunction with state governments, industry, academia, and nonprofits to achieve growth of the US bioeconomy. It also provides context behind what the status of each suggested goal is, factors to consider that may present as challenges to meeting these goals, as well as what research and development investments may be needed to progress, or in some cases to gauge initial feasibility. As stated in the report, "The bold goals set ambitious national targets for the next two decades to help establish R&D priorities that will be critical to advance the bioeconomy. They are not meant to represent commitments by an agency or department to undertake specific activities[2]".

The Plastics Industry Association (PLASTICS) is in support of growing the bioplastics industry and recognizes the vital roles that both bioplastics and traditional plastics play in our daily lives and in development of a circular plastics economy.

As part of the "Climate Change Solutions" section, the Department of Energy includes the goal to: "Spur a Circular Economy for Materials – In 20 years, demonstrate and deploy cost-effective and sustainable routes to convert bio-based feedstocks into recyclable-by-design polymers that can displace more than 90% of today's plastics and other commercial polymers at scale."

Ultimately, The Plastics Industry Association (PLASTICS) is in support of growing the bioplastics industry and

[1] Bold Goals for U.S. Biotechnology and Biomanufacturing | The White House Office of Science and Technology Policy
[2] Bold Goals for U.S. Biotechnology and Biomanufacturing | The White House Office of Science and Technology Policy

recognizes the vital roles that both bioplastics and traditional plastics play in our daily lives and in development of a circular plastics economy. To move in the direction the Bold Goal is proposing, there are a lot of considerations and investments that would need to be made. These include feedstock production, use cases of recyclable biobased plastic, ensuring proper disposal, collection and recycling capacity, testing and certification capacity, and unintended consequences. This document will address what would practically be needed in order to meet the objectives of this goal, as well as key considerations to take into account when working toward it.

Clarifying the Goal

In order to accurately assess what would be needed to reach a goal, it is important to first identify the scope of the goal and metrics for success. The Bold Goal stated above represents a broad idea of replacing >90% of today's plastic with "recyclable-by-design" biobased polymers. However, this goal is too broad, and leaves out necessary specifics. PLASTICS recommends that the following be considered by the Administration to produce a goal more closely aligned with what is feasible and that clearly states the scope of impact.

Beginning of Life – Origin, Feedstocks and Formulation

When discussing polymers synthesized or plastic products produced, the geographic scope needs to be clarified Is the measurement of success based on polymers produced in the United States (US)? While the US is a net exporter of resin, it is essential to note that the US has also imported a significant amount of resin. On average, the US imported $16.6 billion of resin each year between 2013 and 2022. These imported resins are used to create a myriad of plastic items or plastic-containing products. US imports of biobased polymers over the same period averaged $362.2 million, which represented 2.2% of total U.S. resin imports. Are imported resins within the scope of the goal? For the purpose of analysis, this document will focus primarily on US resin production.

As the definition stands today, a biobased bioplastic is a plastic that has some or all of the carbon in the plastic item derived from a renewable source. Therefore, biobased bioplastics can have a varying percentage of biobased content in them. Does this bold goal intend to include plastic products partially derived from biobased content?

Furthermore, it is important to recognize that the polymer is just one component of the final plastic product. Various other substances such as fillers, additives, colorants, etc. are often incorporated into the formulation for a variety of reasons including but not limited to improving strength, durability, flexibility, or changing the color of the plastic. Is the intention to focus exclusively on the polymer component of the finished plastic item, or to include the additional content as well? This document will assume the intention is for 90% of overall polymer production to be sourced from biobased materials.

Lastly, how would the biobased content be measured? The biobased content of a bioplastic can be measured through several ways, and there is no industry consensus on a single universal approach. The most common methods are to measure the biobased carbon content by weight through radiocarbon analysis (ASTM D6866[3]), calculate the biobased content (including carbon, oxygen, nitrogen, and other elements) by weight, or using a mass balance approach, by tracing the flow of biobased feedstock throughout the manufacturing process. The choice of method may depend on the specific characteristics of the bioplastic and the intended application. However, to ensure consistency and transparency, it is essential to specify a preferred method when measuring biobased content of bioplastics.

End of Life – Recycling and Compostability

The Bold Goals reference "recyclable-by-design polymers". While not within the scope of this document, this term is most often associated with the overall design process and is very contingent on the finished product. For example, the Association of Plastic Recyclers organizes their published design guides first by polymer and then by product category. Further, is this definition of "recyclable" meant to include organic, mechanical, and advanced recycling options? PLASTICS believes that any plans to increase circularity should comprise all circular end of life options available, including these three recycling methods stated above as well as any innovative technology that may arise.

For the sake of this assessment, we will assume that the Bold Goal in question is asking for 90% of all plastic polymers only, not necessarily additives, produced in the United States, to be biobased and that "recyclable by design" is intended to include circular end of life options of mechanical, chemical, and organic (commercial composting) recycling.

So, what will it practically take? To answer that question, we first must understand the current state of the US plastics and bioplastics industries.

PLASTICS believes that any plans to increase circularity should comprise all circular end of life options available, including organic, mechanical, and advanced recycling methods as well as any innovative technology that may arise.

[3] ASTM D6866 "Standard Test Methods for Determining the Biobased Content of Solid, Liquid, and Gaseous Samples Using Radiocarbon Analysis" | American Society for Testing and Materials

Understanding the Desired Degree of Change – What Does the Bioplastics Market Look Like Today?

Polymer Types

There are hundreds of polymers, each having unique characteristics that are important in the applications where used. Of those, the 5 most common polymers are: Polyethylene terephthalate (PET), Polyethylene (PE), High-density polyethylene (HDPE), Low-density polyethylene (LDPE), Polypropylene (PP), and Polyvinyl chloride (PVC)[4]. Only PE is currently commercially available as 100% biobased, though partially biobased PET is available, and biobased PP is under development. While it is possible to make PVC completely biobased, it is not commercially available due to cost considerations. Appendix 1 holds a table of the current common bioplastics either produced, in development to commercialize, or are popular targets for research.

Biobased precursors are required to create biobased polymers. For example, PE is produced by condensation polymerization of ethylene gas. Biobased ethylene gas is produced through dehydration of ethanol from sources such as sugar cane[5]. Not all polymers produced currently have biobased precursors commercially available. In 2013, the Department of Energy created a list of the top 30 chemicals from biomass that can be "converted to a number of high-value bio-based chemicals or materials". Of the candidates identified that can be used to produce plastics, the table in Appendix 2 highlights those which are commercially available at this time[6]. Future research and investment into the commercialization of the other biobased precursors will be needed to ensure the broadest number of biobased polymers available.

Market Size

Due to the relatively small size of the industry, obtaining robust and comprehensive global data on the bioplastics market can be challenging. European Bioplastics published in 2022 a report that showed 2.22 million metric tons, or 4.89 billion pounds, of bioplastics were globally produced in 2021. However, this measurement would also include a small amount of biodegradable plastics that are not biobased. This is estimated to be less than 4% of overall bioplastics production. Further, this number does not include biobased polyurethanes which can be a large amount of overall bioplastic production. The US is estimated to have 18.9% of the total global production capacity, after Asia (41.4%) and Europe (26.5%). This converts to approximately 0.89 billion pounds of US produced biobased resin. This number must be

[4] The Resin Review – 2021 | American Chemistry Council
[5] I'm Green™ Bio-Based | Braskem
[6] Top Value Added Chemicals from Biomass | U.S. Department of Energy

extrapolated from global estimates as there is not currently a standard measurement method for the bioplastic industry segment in U.S. NAICS or NAPCS codes.

In 2022, 125.5 billion pounds of resin was produced in the United States[7]. Therefore, it is a reasonable baseline estimate that biobased resin production would account for 0.71% of the overall US plastic resin production in 2022. To reach 90% market share by 2043, the compound annual growth rate would need to be more than 27%. While estimates vary among market research firms, the estimated CAGR for recent time periods are well below that rate. For example, *Global Biopolymer and Bioplastic Market 2023-2027* forecasts the CAGR over the next 5 years to be 14.36%.[8]

Biobased resin production would account for 0.71% of the overall US plastic resin production in 2022. To reach 90% market share by 2043, the compound annual growth rate would need to be more than 27%

Reaching this growth rate is a rather “bold” goal and more aspirational than other country’s targets. The European Commission states in their *Communication from the Commission to the European Parliament and the Council on ‘Sustainable Carbon Cycles’* an objective to achieve at least 20% (current level is 10%) of the carbon used in chemical and plastic products be from sustainable non-fossil resources by 2030[9].

Feedstock Production

Bioplastics sources are categorized as first, second, and third generation feedstocks[10].

- First generation refers to food crops such as sugar cane, corn, and soybeans.
- Second generation includes nonfood crops such as cellulose from wood and by-products generated from food crops, such as corn husks or waste vegetable oil.
- Third generation feedstocks are biomass derived from algae.

There is often concern about the amount of land-use associated with bioplastic production as it relates to food availability. However, even with a drastic increase in bioplastic production, food availability will not be threatened[11]. In 2022, global production of bioplastics used 0.015%

[7] 2022 Resin Situation and Trends | American Chemistry Council
[8] Global Biopolymers And Bioplastic Market 2023-2027 | Report Linker
[9] Sustainable Carbon Cycles (pg. 17) | European Parliament
[10] Bioplastic Feedstock 1st, 2nd and 3rd Generations | Bioplastics News
[11] Frequently Asked Questions | European Bioplastics

of arable land. For comparison, food and animal feed is currently grown on 73% of global arable land[12].

It is possible to create a rough estimate of the amount of arable land that would be necessary to use in the United States. Using the global production estimate above where 0.015% of all arable land globally are used to produce bioplastic feedstocks, the amount of US arable land needed to grow renewable feedstocks for 90% of US plastic production would be .44%[13]. This is an over estimate which assumes that all feedstock would come from first generation sources and an average global productivity per acre, recognizing that the US has a higher productivity per acre rate than most other countries.

While first generation feedstocks are a viable resource when not needed for food or feed, bioplastics are increasingly being derived from non-food sources. Second and third generation feedstocks provide a significant growth potential for increasing bioplastic production and beneficial sustainability attributes.

> The amount of US arable land needed to grow renewable feedstocks for 90% of US plastic production would be .44%

Utilization of nonedible byproducts from first generation feedstocks provides another source of income for farmers. The addition of this viable market to a farming business could lead to further stabilization of food prices as there would be higher security of demand for all parts of the crop. Using scraps for bioplastic production captures the carbon sequestered by the plant during its growth phase into a stable form.

Strengthening markets for non-food crops (second gen) and algae (third gen) would lead to responsibly managed farms and innovations in these spaces to increase production capacity and efficiency. As the demand for this material increases, the market will organically respond by increasing the production capacity of establishments such as sustainably managed forests, seaweed farms, and algae blooms. Innovations in processing these feedstocks into bioplastics, such as in biorefineries, would be incentivized in tandem.

In order to reach the capacity needed for 90% of polymers to be biobased, however, government support must be provided to feedstock and raw material producers as well as product and resin manufacturers to spark this higher demand**.** Pricing of biobased raw materials is typically higher than fossil fuels. It is important to note that subsidies for biobased raw materials for fuels do not positively impact the pricing of raw materials for plastic products.

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[12] Renewable feedstock | European Bioplastics
[13] See Appendix 3 for equation used.

The development of biobased plastic feedstocks has the potential to be a viable contribution to the US economy and drastically reduce greenhouse gas emissions in the production of plastic. However, this would not be without major implications for the current plastics industry. The following logistic and life cycle considerations should be taken into account.

- What will the socioeconomic impact be on communities that currently rely on natural gas extraction for the production of plastics? Likewise on communities relying on refining facilities.
- As the need for biorefineries or other technologies for processing biobased feedstocks into plastics increases, what is the impact of building these? Would existing refineries be retrofitted?
    - Understanding that the locations of current feedstocks (mainly natural gas in the US) and biofeedstocks (from land farms, seaweed farms, algae blooms, etc.) would be different, what would the impact be on distributing biofeedstocks to retrofitted or new biorefineries? Then to processors/converters?
- Once biobased resins get to processors/converters, are these facilities currently equipped to convert this material into bioplastic products? Would facilities currently processing fossil-based resins need to be retrofitted?

Use Cases of Recyclable Biobased Plastic

Biobased bioplastics are currently used in many of the applications that fossil-based plastics are. The numerous biobased polymer types that exist provide various degrees of durability, flexibility, heat resistance, etc. that make them functional for a wide range of applications. Today, bioplastics can be found in the following market segments[14]:

- Packaging
- Food services
- Agriculture & horticulture
- Consumer electronics
- Automotive & transport
- Consumer goods and household appliances
- Building & construction
- Coating & adhesives
- Fibers

[14] Applications for bioplastics | European Bioplastics

However, there are stipulations for some of these segments that may make it more difficult to use biobased plastic as a drop-in replacement. For instance, food and medical packaging must meet stringent standards to ensure that the products they are enclosing are protected. Agricultural products have a higher variability of toxicity dependent on the environment in which they were grown as well as chemicals such as fertilizers and pesticides used during the growth stage. Food and medical packaging manufacturers have expressed concern for the transition to biobased in these applications for anticipation of increased regulation and certification of polymers, which may cause supply delays or shortages, and the potential harm caused to the products these packages are in contact with, resulting in potential harm to humans upon consumption/use.

Benefits can be realized if negligible toxicity can be guaranteed in food packaging. Opportunity exists for more food and organics waste to be diverted from landfill which would also help to reach the second bold goal within the Food & Agriculture category - "reducing methane emissions from food waste in landfills, to support the U.S. goal of reducing greenhouse gas emissions by 50%"[15].

> Compostable bioplastics can enable more food and organics waste to be diverted from landfill. This would also help to reach the second bold goal within the Food & Agriculture category.

Another concern for manufacturers is the higher price of biobased plastics. Price premiums anywhere from 10% to 200% are often seen with biopolymers such as PLA, BioPET, BioPP, etc. Reports have shown that consumers are willing to pay a price premium for more sustainable options, often referred to as the "green premium". Though, this willingness drastically drops once the green premium exceeds 20%[16]. This issue is not unique to biobased content. Similar challenges are faced by other materials with environmental attributes, such as recycled content.

Bringing awareness to the concerns of some industry segments will initiate increased innovation and problem solving in the biobased plastic industry. However, these solutions will require monetary and temporal investment.

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[15] Bold Goals for U.S. Biotechnology and Biomanufacturing pg. 17 | The White House Office of Science and Technology Policy

[16] "GreenPremium prices along the value chain of bio-based products" | Renewable Carbon Publications

Ensuring Circularity – Consumer Behavior

Proper disposal is essential to closing the loop of the circular economy. Many factors play into whether a product is recycled or composted including human behavior, product labeling, consumer education, infrastructure availability and effectiveness, and market demand for the compost or recycled content.

Circular waste management is a complex system. One of the largest barriers is human behavior which is rooted in societal/cultural norms and expectations and is very difficult to change. There are also organizations such as GreenPeace[17] [18] who are actively advising against plastic recycling which is demeaning consumer's faith in the system.

In order to ensure that consumers are properly disposing (via organic, mechanical, or advanced recycling), of "recyclable-by-design" biobased plastics, large investments in product labeling, consumer education, and availability of disposal receptacles are needed. The answer to the question "what do I do with this product once I'm done with it?" must be intuitive to understand and convenient to perform. It is also essential that consumers disassociate "biobased" with soil or marine "biodegradable" to avoid instances of littering.

Still, no matter how clear the communication is to the consumer, mistakes are inevitable; products intended for recycling may end up in the compost facility and vice versa. There are ways to prepare for these contaminants to make sorting them out easier for each end-of-life facility. For instance, it is difficult for a composter to pick out non-compostable plastic contaminants if they look similar to compostable plastics. The certification label or RIC is often difficult to see once co-mingled with food and yard waste. Full product tinting is a potential solution to this challenge. If all certified compostable plastics were tinted green and all non-compostable plastics were restricted from using green tinting, it would be easier for composters to distinguish between them.

This same tinting would be useful for sorting out compostable plastics within a Materials Recovery Facility (MRF). Further measures for sorting between polymer types at MRFs such as polymer labeling, and optical sorting equipment would be necessary investments as well.

Ensuring Circularity - Recycling and Composting Capacity

Regardless of feedstock, more recycling and composting facilities are needed in order to handle the amount of plastic material being disposed of today. The annual U.S. mechanical

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[17] Plastics Industry Association Response to Greenpeace Report, Attacks on Plastic Recycling | Plastics Industry Association
[18] Circular Claims Fall Flat Again | Greenpeace USA

recycling capacity for plastic in 2021 was 6.72 million tons[19]. About 45 advanced recycling facilities exist in the U.S with a combined capacity to handle over 950 thousand tons annually. However, only about 15% of these facilities are currently operational. ICIS estimates that capacity of advanced recycling technologies will reach 5 million tons by 2025[20]. According to GreenBlue, of the 964 composting facilities in the US, 107 (11%) of them accept compostable bioplastics[21]. The volume of compostable plastic processed by these facilities is unknown.

There is no possibility of reaching a truly circular economy without sufficient disposal capacity available. Government and industry support for increasing the capacity of each end-of-life option through current technology evolution and expansion as well as new technology development is essential. In the last 5 years, the plastics industry has invested approximately $1.7 billion in mechanical recycling alone. PLASTICS is working on an updated Bioplastics Market Watch Report that will include estimations of similar investments from industry for circular bioplastic disposal.

It is important to note how biobased plastics are currently accepted and processed through mechanical recycling streams. Some biobased plastics, such as BioPP and BioPET are, for all intents and purposes, the same polymer as their fossil-based counterparts (PP, PET) and can be recycled right alongside these streams. These bioplastics have the same property and processing characteristics as their fossil-based equivalents because those properties are driven by the polymer chemical structure rather than by the source of the carbon.

Other recyclable biobased polymers that do not have a fossil-based equivalent would need to be sorted separately at the MRF. While this is technically possible, it can be a large financial investment for the MRF to add another sortation stream. Separate sorting of biobased polymers is not yet common because the volume of these polymers is not high enough to produce profitable bales or a steady end market to sell to. This variability may not justify the cost of new equipment needed to add this sortation capability. Therefore, recyclable biobased polymers often are discarded or viewed as contaminants to the other plastic recycling streams.

There is no possibility of reaching a truly circular economy without sufficient disposal capacity available. Government and industry support for increasing the capacity of each end-of-life option through current technology evolution and new technology development is essential.

[19] How the US can achieve high plastic recycling rates | I.C.I.S.
[20] Recycling Supply Tracker – Chemical, 2021 | I.C.I.S.
[21] Mapping Composting Infrastructure and Supporting Legislation | GreenBlue

Adequate funding for MRFs would alleviate this barrier to sorting for biobased recyclable polymers.

For plastic to be organically recycled, it must be certified as commercially compostable. This means that the plastic is designed to be composted in a municipal or industrial aerobic composting facility, meets ASTM D6400, D6868, or an equivalent standard, and is recognized by an independent 3rd party certification body, such as the Biodegradable Products Institute (BPI). As stated above, the majority of industrial composting facilities in the US do not currently accept compostable bioplastics. Though this is only one end-of-life option for bioplastics, it is important to increase the acceptance rate to allow for ease of proper disposal of compostable bioplastics. Composters could be rejecting these materials due to numerous factors including potential for contamination of non-compostable plastics, inability to process compostable plastics in a desired amount of time, or inability to market compost as organic if it contains compostable plastic.

It is certainly possible to run an efficient and profitable composting facility while accepting compostable plastics, though. Composters that do accept certified commercially compostable plastic are able to process the material effectively at their facilities and have noted that these bioplastics are often the first to breakdown. In order to increase capacity, more commercial compost facilities would need to accept certified compostable plastics. This would require incentives for composters to adjust their processes to ensure effective decomposition, assurance that their compost could be sold as organic, and a more effective sortation system for keeping contaminants out.

Testing and Certification Capacity

Lastly, capacity for testing and certification at both the beginning and end of life of biobased plastic products should be taken into consideration. At the beginning of the product's life, quantification and certification of biobased content percentage would be necessary for accurate measuring and labeling.

The USDA BioPreferred Program tests and certifies biobased content in products. Their process, from application through certification, takes at least 90 days. Each new product desiring biobased content certification must go through this process. If a product's formulation changes by more than 3% (+/-) biobased content, it must be retested in order to redetermine biobased content percentage. Considering that the US would have about 89% more biobased products in the US if this Bold Goal was reached, how many more testing facilities would be needed to handle this volume?

The cost of certification must also be considered. Currently, "Companies are responsible for the cost of biobased content testing … and periodic re-testing during audits. Product testing is completed using ASTM D6866, which costs approximately $400 USD per sample, plus shipping costs."[22] This additional cost could be a deterrent for companies not currently manufacturing biobased products to want to make the switch.

As stated above, a viable circular end of life option for some bioplastics is to commercially compost them. However, in order to be accepted into a composting facility, the product must be certified as commercially compostable. Similar investments would have to be made in order to handle a large increase in products needing these testing and certification services. BPI is the most common certification body of this kind in the US. Currently, their process from application, to testing, to certification, takes a minimum of 6 months and pricing starts at $1,500 per product. The testing required is a lab test. Some companies opt into conducting field testing as well to ensure biodegradation in numerous composting processes and conditions. Field testing would require an additional monetary and time investment. BPI certifications are valid for 3 years, at which point recertification is required, costing $1,000 per product.

Recommendations for Next Steps

The *Bold Goals for US Biotechnology and Biomanufacturing* propose the goal of producing more than 90% of today's plastic and other commercial polymers with recyclable-by-design biobased polymers by 2043. With the current US biobased plastic production capacity of 0.71%, PLASTICS believes that this goal is exceedingly aspirational as proposed. PLASTICS supports the idea of diversifying feedstocks and increasing biobased plastic production and improving circular end of life options. To move in this direction, PLASTICS recommends the following.

1. Use existing programs such as the Farm Bill and other existing or new rural development programs to incentivize renewable feedstock production as well as further research and development to introduce new feedstocks as well as growth efficiency.
2. Set "sustainable content" limits that accept biobased content in addition to recycled content minimums to support both industries. Minimum recycled content policies support the development of recycling supply chains, however in the short term can disincentivize the use of biobased content in a product as each has a premium over virgin fossil-sourced plastic.
3. Develop NAICS and NAPCS codes for bioplastic product manufacturing in order to properly enable characterization of the economic value of this segment of the U.S. bioeconomy. Utilizing codes specific to bioplastics would greatly enhance the ability for

[22] Frequently Asked Questions | USDA BioPreferred Program

policymakers and other stakeholders to make more informed decisions. Measuring growth in economic areas like jobs and average wages is key to understanding how public policy is helping this new industry, what barriers should be addressed, and where investment is needed[23].

4. Encourage development of biobased additives in addition to polymers to facilitate products with higher biobased content percentages overall.
5. Promote preferential purchasing programs such as USDA BioPreferred Program that require the federal government to purchase biobased products. This incentivizes the development of biobased plastic product production.
6. Consider providing fee reductions for biobased products such as in extended producer responsibility (EPR) programs where fees are set based on recycling rates of products. Those with higher recycling rates often have lower associated fees.
7. Develop new programs to provide loans, grants, and other incentivization to provide further opportunities for bioplastic industry growth.
    a. Provide aid for producers of products that are associated with desirable organic wastes, such as food contact packaging, cutlery, mulch films, etc. and are not better fit for recycling, to certify their products as commercially compostable.
    b. Extend incentives that exist for biofuels to biomaterials as well.
8. Leverage existing policy to support expansion of all circular end of life options, including both recycling and composting.
    a. The RECOVER Act helps to increase collection capabilities of recyclable and compostable plastics, including residential and commercial access.
    b. The RECYCLE Act stimulates investments in consumer education to ensure comprehension of how to properly dispose of different products.
    c. The Plastic Waste Reduction and Recycling Act coordinates the continuous development and improvement of new recycling technologies.
9. Create new policies or programs to further support and provide opportunities for growth of recyclers and composters.
    a. Develop new programs or policies to subsidize investments in sortation and reprocessing capabilities at MRFs, recyclers, and composting facilities. EPR strategies could enable funding for this infrastructure.

[23] 2023 Farm Bill Priorities | Plant Based Product Council

b. Incentivize sending organic waste, including compostable plastic to commercial composters, anaerobic digesters, or other available facilities instead of sending to landfill. In tandem, ensure that the addition of this material stream can be handled and feasibly processed by these facilities without sacrificing the quality of their end product.

10. Develop a comprehensive labeling requirement to differentiate recyclable and compostable plastics and clearly signify preferred end of life disposal method to the consumer and end of life facility. Consider adopting recommendations from BPI[24] and US Composting Council[25].
11. Recognize biobased content measurement methods that align with industry needs and enable the introduction of more biobased feedstocks into existing processes. Whether using the biomass balance or carbon-14 approach, ensure that the biobased content amount and measurement method is qualified.
12. Consider externalities in all decision-making processes. For instance, socioeconomic impacts on communities that may be affected by any policies or programs or environmental impacts of changes in supply chain logistics.

> The intent of the Bold Goals is to increase the amount of chemicals and plastic products made with biobased feedstocks, as well as to have circular end of life options available. PLASTICS believes that increasing feedstock diversity for plastic production and circular disposal options will positively contribute to a sustainable, circular economy.

Conclusion

The main societal impacts these Bold Goals are aimed at addressing are climate change solutions, food and agricultural innovation, supply chain resilience, human health, and job growth. It is noted that the Bold Goals do not use language or definitions consistently, however, it seems that the intent is clearly to increase the amount of chemicals and plastic products made with biobased feedstocks, as well as to have circular end of life options available. PLASTICS believes that increasing feedstock diversity for plastic production and circular disposal options will certainly help to make these positive impacts a reality.

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[24] Labeling | Biodegradable Products Institute
[25] Home Page | US Composting Council

While the bioplastics industry is expected to grow at an annual rate of 14%, the >27% growth rate needed to achieve 90% biobased by 2043 is not something that the market will organically achieve. To enable a substantial increase in biobased plastics being produced, investments must be made in raw material and feedstock development, supply chain logistics, manufacturing capability, recycling and composting capacity and effectiveness, and testing and certification capacity.

Appendix 1

Polymer Abbreviation	Polymer Name	Biobased[26]	Biodegradable[27]	Certified Products Available[28]
C	Cellulose	Yes	Yes	A, B, C, D, E, F
CA	Cellulose Acetate	Yes	Yes	A, B, C, D, E, F
PA 10	Polyamide 10	Partially	No	
PA 1010	Polyamide 1010	Yes	No	
PA 11	Polyamide 11	Partially	No	
PA 410	Polyamide 410	Partially	No	
PA 610	Polyamide 610	Partially	No	
PBAS	Polybutylene adipate-co-succinate	In development	Yes	
PBAT (adipic acid)	Polybutylene adipate-co-terephthalate	No	Yes	A, B, C
PBAT (azelaic acid)	Polybutylene azaeli-co-terephthalate	Partially	Yes	A, B, C, D
PBS	Polybutylene succinate	Yes	Yes	A, C, D
PBSA	Poly(butylene succinate-co-butylene adipate)	Partially	Yes	A, B, C
PCL	Polycaprolactone	Yes	Yes	A, B
PDS	Polydioxanone		yes	
PE	Polyethylene	Yes	No	
PEF	Polyethylene furanoate	In development	No	
PES	Polyethylene succinate	Partially and fully biobased in development	Yes	A, C, D
PET	Polyethylene terephthalate	Partially	No	
PGA	Polyglycolide or Polyglycolic acid	No	Yes	A
PHA	Polyhydroxy alkanoate	Yes	Yes	A, B, C, D, E
PHB and copolymers[29]	polyhydroxybutyrate	Yes	Yes	A, B, C, D, E, F
PLA	Polylactic acid	Yes	Yes	A, E
PLG or PLGA	Poly(lactic-co-glycolic acid)	Yes	Yes	
PP	Polypropylene	Yes	No	

[26] Denotes product types that have biobased grades available. All grades of this polymer type may not be biobased.
[27] Biodegradability is either established by academic research and/or claims by material suppliers.
[28] A (industrially compostable); B (home compostable by Vinçotte OK Home certification); C (soil biodegradable); D (marine biodegradable); E (anaerobically digestible); F (freshwater biodegradable). Grades meeting certain forms of biodegradability are supported with industry consensus standards where available. Not all grades of each polymer type may meet the form of biodegradability noted; please check with the material supplier for the biodegradability of specific bioplastic grades.
[29] Including P_3HB, P_4HB, P_3HB_4HB, P_3HB_3HV, $P_3HB_3HV_4HV$, P_3HB3_3Hx, P_3HB_3HO, P_3HB_3HD

PPA	Polyphthalamide	Partially	No	
PPE	Polyphenylene ether	Partially	No	
PPF	Polypropylene fumarate	No	Yes	
PTF	Polytrimethylene furandicarboxylate	In development	No	
PTT	Polytrimethlene terephthalate	Partially	No	
TPC-ET (TPC or COPE)	Thermoplastic Copolymer Elastomer	Partially	No	
TPS	Thermoplastic starch	Yes	Yes	A, B, C, D, E, F
TPU	Thermoplastic polyurethane	Partially	No	

Appendix 2

Building block	Precursor / Derivative	Potential Uses	Transformation Pathway
Four Carbon 1,4-Diacids (Succinic, Fumaric, and Malic)	butanediol (BDO)	solvents and fibers	reductions
	tetrahydrofuran (THF)		
	gamma-butyrolactone (GBL)		
	pyrrolidinone family	water soluble polymers	reductive aminations
2,5-Furan dicarboxylic Acid (FDCA)	2,5-bis(aminomethyl)- tetrahydrofuran	polyesters and nylons	reduction
	2,5-dihydroxymethylfuran	polyesters	direct polymerization
	2,5-bis(hydroxymethyl)tetrahydrofuran		
3-Hydroxypropionic Acid (3-HPA)	acrylic acid	super absorbent polymers	dehydration
	1, 3 propane diol	sorona fiber	reductions
Glucaric Acid	polyhydroxypolyamides	nylons through amination	direct polymerization
Glutamic Acid	1,5-propanediol	monomers for polyesters and polyamides	reductions
	5-amino, 1-butanol		
Itaconic Acid	2-methyl-1,4-BDO	solvents and fibers	reductions
	3-methyl THF		
Levulinic Acid	b-acetylacrylic acid	copolymerization with other monomers for property enhancement	oxidation
	diphenolic acid	replacement for bisphenol A used in polycarbonate synthesis	condensation
Glycerol	glyceric acid	PLA with better polymeric properties, polyester fibers with new properties	oxidation
	1,3-propanediol	sorona fiber	bond breaking (hydrogenolysis)
	branched polyesters	unsaturated polyurethane resins for use in insulation	direct polymerization
Sorbitol (Alcohol Sugar of Glucose)	isosorbide	PET like polymers such as Polyethylene isosorbide terephthalates	dehydration
	2,5-anhydrosugars		
	propylene glycol	PLA	bond cleavage (hydrogenolysis)
	lactic acid		
Xylitol/arabinitol (Sugar alcohols from xylose and arabinose)	xylaric acid	new polymer opportunities	direct polymerization

Appendix 3

$$.015\%\ global\ arable\ land\ needed\ for\ current\ production \div 4.89\ billion\ lbs = X\%\ US\ arable\ land\ needed\ for\ current\ produciton \div 0.89\ billion\ lbs$$

X = 0.00345% of US land needed for current production

125.5 billion lbs. of resin produced in US total

90% of this = 112.95 billion lbs.

$$0.00345\%\ US\ land\ for\ current\ capacity\ \div 0.89\ billion\ lbs\ (current\ capacity) = X\%\ land\ needed\ for\ 90\% \div 112.95\ billion\ lbs\ (90\%\ capacity)$$

X = 0.44% of US arable land needed for 90%, or 112.95 billion lbs. of resin